

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Agentic AI-Based Smart Compliance Auditor & AI Security Validation Platform for Enterprise LLM Systems

The rapid adoption of Artificial Intelligence (AI) and Large Language Model (LLM)-based applications in enterprises has introduced new cybersecurity challenges alongside increasing compliance requirements. Organizations must comply with security standards such as ISO/IEC 27001 and the NIST Cybersecurity Framework while protecting AI systems from threats such as prompt injection, jailbreak attacks, and sensitive data leakage. Although existing compliance management and AI security tools are available, they typically address these problems independently and lack an integrated framework that combines intelligent compliance auditing, AI security validation, collaborative multi-agent reasoning, and automated remediation support.

This research proposes an **Agentic AI-Based Smart Compliance Auditor & AI Security Validation Platform for Enterprise LLM Systems**, an intelligent web-based platform that integrates cybersecurity compliance assessment with automated AI security testing through a collaborative multi-agent architecture. The proposed system evaluates organizational security controls against simplified ISO/IEC 27001 and NIST requirements while also performing automated AI security validation by simulating prompt injection, jailbreak, privilege escalation, and sensitive information disclosure attacks against enterprise AI applications.

The framework employs multiple specialized AI agents, including Policy, Compliance, Risk, Red Team, Blue Team, Judge, Debate, Decision, Forecast, Roadmap, Memory, and Report Agents. These agents collaboratively analyze compliance, classify vulnerabilities based on the OWASP Top 10 for LLM Applications, generate remediation recommendations, predict future compliance scores, compare previous audit results, and produce business-oriented security summaries. An interactive dashboard presents compliance scores, AI security findings, remediation roadmaps, business impact analysis, and downloadable audit reports to support effective cybersecurity decision-making.

The framework is implemented using Python, FastAPI, Streamlit, SQLite, and Google Gemini API with an offline rule-based fallback, providing a scalable and user-friendly platform for enterprise AI security assessment. The proposed research contributes by integrating compliance auditing, AI security validation, intelligent decision support, and automated remediation planning into a unified multi-agent cybersecurity platform, which can be extended in the future with real-time monitoring, CI/CD integration, predictive analytics, and advanced AI governance capabilities.

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References (Category specific)

1.

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature

Project Guide

Name:

Signature:

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.